

A Machine Learning Approach to Speaker Age Prediction from Audio and Metadata

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Abstract

This report studies a supervised regression task in which speaker age is predicted from structured speech-related inputs, including metadata, extracted audio features, pitch and voice-quality measures, and speech-duration descriptors. Three models were evaluated: a baseline DummyRegressor with a mean strategy, ExtraTrees, and HistGradientBoosting. Model comparison under cross-validation shows that the machine learning approaches clearly outperform the naive baseline, indicating that the available features contain meaningful age-predictive signal. HistGradientBoosting achieved the strongest initial result and was therefore selected for hyperparameter tuning. After optimization, the tuned HistGradientBoosting model obtained a test RMSE of approximately 9.04 years, substantially better than the baseline test RMSE of approximately 12.92 years. Additional feature-group experiments show that the best performance is obtained when all feature types are combined, while diagnostic plots indicate that prediction is more reliable for younger speakers than for older speakers. The main limitation is dataset imbalance: the age distribution is concentrated among younger speakers, which weakens reliability at older ages.

1 Problem Overview

The goal of this project is to predict speaker age from structured features derived from speech and accompanying metadata. Automatic age estimation from speech has been studied before, which supports the idea that speech carries age-related information.[5] This is a regression problem because the target variable, age, is continuous rather than categorical. The available predictors cover several complementary aspects of speech production, including pitch, silence and pause behavior, duration, lexical volume, and voice quality, together with metadata variables. The

Table 1: Dataset summary statistics.

Statistic	Value
Number of samples	2933
Number of columns	20
Minimum age	6
Maximum age	97
Mean age	approximately 27.9
Median age	23.0
Age standard deviation	approximately 13.09

main technical challenge is to build a model that predicts age accurately from several different types of features. These feature groups may contribute unequally, and they may also interact in nonlinear ways.

2 Dataset and Exploratory Analysis

The dataset contains 2,933 samples and 20 original columns, with speaker ages ranging from 6 to 97 years (Table 1). The mean age is approximately 27.9 years and the median is 23.0 years, already indicating that the sample is concentrated toward younger speakers. Figure 1 confirms that the age distribution is strongly right-skewed: younger speakers are much more frequent, whereas older speakers appear relatively rarely. This matters for the later results. A model trained on this dataset is likely to work best where many examples are available and less well at the edges of the age range.

The correlation heatmap in Figure 2 shows the main structure in the feature space. Some variables are strongly correlated, such as `num_words` and `num_characters`. Pause- and silence-related variables also show links with age, suggesting that nonlinear models may be useful.

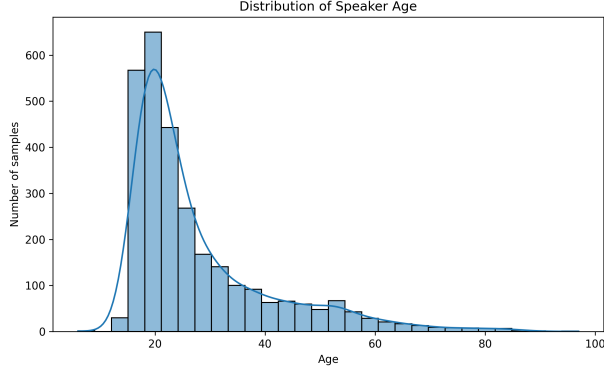


Figure 1: Age distribution of the dataset.

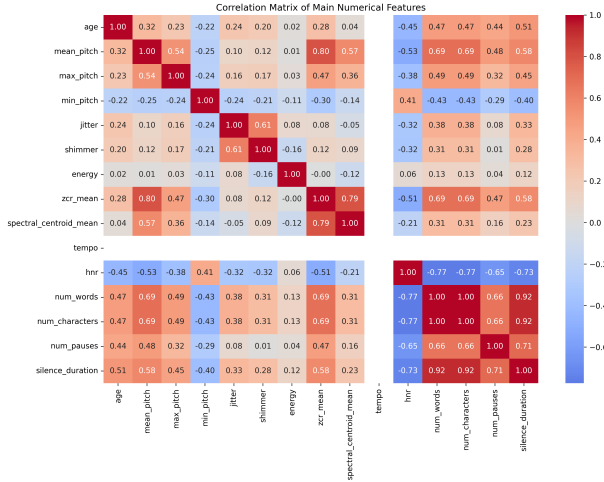


Figure 2: Correlation heatmap for selected features.

3 Methodology

3.1 Feature Sets

The experiments were designed to answer two questions. First, which model performs best? Second, which groups of features contribute most to age prediction? Four feature configurations were compared: all available features; extracted audio features only; provided CSV or metadata features only; and pitch plus voice-quality features only. This design helps compare the full feature set with smaller groups of features.

3.2 Models

Three regression approaches were considered. The baseline model is DummyRegressor with a mean strategy. It predicts the average training-set age for every example and serves as a naive reference point. Its role is important because any useful machine learn-

ing model should improve clearly over such a trivial predictor. ExtraTrees is an ensemble of randomized decision trees. It can capture nonlinear relationships and feature interactions, and it often works well on tabular data.[2] HistGradientBoosting is a gradient-boosted tree model designed for tabular regression. It can learn nonlinear patterns efficiently and is practical for a dataset of this size. In scikit-learn, this model uses a histogram-based implementation, and the official documentation notes that it is inspired by LightGBM.[4, 3] Its underlying idea comes from gradient boosting methods for additive tree models.[1]

3.3 Evaluation Metric

Performance was evaluated using RMSE. Since age is continuous, RMSE is appropriate and reports error directly in years. Lower RMSE indicates better performance.

3.4 Cross-Validation

Cross-validation was used to avoid relying on a single train-test split. Averaging results across folds gives a more reliable comparison and shows how stable each model is.

3.5 Hyperparameter Tuning

After the initial comparison, the best-performing model was selected for hyperparameter tuning in order to improve generalization and reduce prediction error beyond the untuned configuration. The results show that HistGradientBoosting was selected for tuning and remained the strongest model after optimization. The best configuration combined `min_samples_leaf = 50`, `max_leaf_nodes = 63`, `max_iter = 500`, `max_bins = 128`, `learning_rate = 0.03`, and `l2_regularization = 1.0`. More importantly, tuning improved the best cross-validation RMSE to approximately 10.00 and also gave the best final test result. This shows that tuning produced a real performance gain rather than simply keeping the same model ranking.

4 Model Comparison and Final Results

Table 2 and Figure 3 show the initial cross-validation results. HistGradientBoosting achieved the best mean

Table 2: Cross-validation model comparison.

Model	CV RMSE	CV RMSE Std.
HistGradientBoosting	approximately 10.20	approximately 0.58
ExtraTrees	approximately 10.42	approximately 0.70
DummyRegressor (mean)	approximately 13.12	approximately 0.73

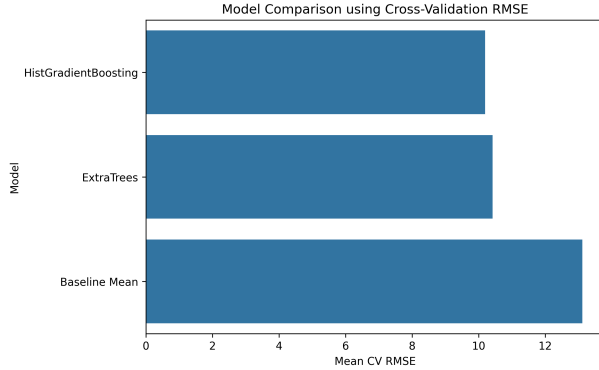


Figure 3: Cross-validation RMSE across models.

CV RMSE at approximately 10.20 years, with a fold-to-fold standard deviation of approximately 0.58. ExtraTrees was close behind at approximately 10.42 years, with a standard deviation of approximately 0.70. Both machine learning models performed substantially better than the baseline DummyRegressor, whose RMSE was approximately 13.12 years. This gap is the main result of the initial comparison. It shows that the speech-derived features and metadata contain useful information for age prediction, and that nonlinear ensemble models can use this information effectively.

The strong performance of both HistGradientBoosting and ExtraTrees suggests that tree-based ensemble models are well suited to this task. These models can learn nonlinear relationships between speech features, metadata, and age. Since HistGradientBoosting achieved the lowest RMSE and slightly more stable cross-validation results, it was selected for hyperparameter tuning.

The final tuned-model results support the same conclusion. As shown in Table 3, HistGradientBoosting remained the best model after tuning. Its best cross-validation RMSE after tuning improved to approximately 10.00 years. The final test RMSE was approximately 9.04 years, and the test MSE was approximately 81.69. The baseline test RMSE was approximately 12.92 years, so the tuned model reduces held-out error by nearly four years relative to the naive reference. This is a clear improvement in practice. It suggests that the model is learning useful patterns instead of only fitting noise in the training data.

It is also useful to compare the tuned cross-validation result with the final test result. The test RMSE is even lower than the tuned cross-validation RMSE. This suggests that the selected model generalizes well to the held-out test set. This does not remove concerns about class imbalance or subgroup performance. However, it does support the conclusion that HistGradientBoosting is the strongest overall approach among the tested models. Overall, the modeling story is clear. HistGradientBoosting performs best in the initial comparison, improves further after tuning, and achieves the best final test accuracy.

Table 3: Summary of final tuned-model performance.

Metric	Value
Best model before tuning	HistGradientBoosting
Baseline model	DummyRegressor (mean)
Best CV RMSE after tuning	approximately 10.00
Final test RMSE	approximately 9.04
Final test MSE	approximately 81.69
Baseline test RMSE	approximately 12.92

5 Feature Analysis

After identifying the best model, feature analysis was used to understand which feature groups and individual variables contributed most to age prediction.

5.1 Feature Group Comparison

Table 4 and Figure 4 compare the four feature sets. The full feature set gave the best result, with a cross-validation RMSE of approximately 10.20 years. Extracted audio features alone were only slightly worse at approximately 10.57. This suggests that they contain a large part of the useful predictive signal. Provided CSV or metadata features alone achieved approximately 10.87. This is still clearly better than the baseline, but weaker than the richer feature sets. Pitch and voice-quality features alone performed worst, with an RMSE of approximately 11.93.

These findings help explain the task more clearly. First, no single feature family is enough on its own because the best result is achieved when all information is combined. Second, extracted audio descriptors are especially useful because they perform almost as well as the full feature set even on their own. Third, the weaker performance of the pitch and voice-quality subset shows that age prediction cannot be explained by a small set of vocal measures alone. Instead, the best results come from combining acoustic, temporal, and metadata information. This suggests that age-related

Table 4: Feature-group comparison.

Feature group	N	CV RMSE
All features	295	approximately 10.20
Extracted audio features only	111	approximately 10.57
Provided CSV / metadata features only	184	approximately 10.87
Pitch / voice-quality features	15	approximately 11.93

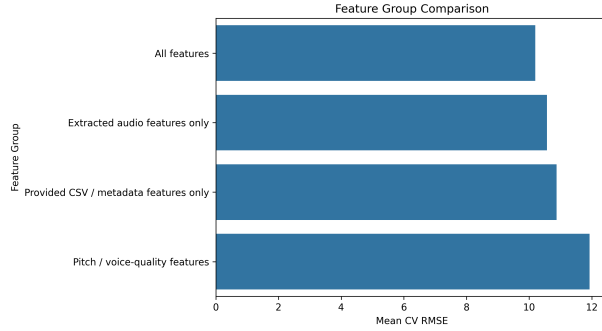


Figure 4: Prediction accuracy by feature group.

signal is spread across several aspects of speech rather than concentrated in a single simple cue.

5.2 Individual Feature Importance

Permutation importance was used to examine which variables contributed most to the final model’s predictions. As shown in Table 5 and Figure 5, **aud_duration_audio** is the most influential feature, with a mean importance of approximately 1.56. This indicates that the duration of the audio recording provides a strong signal for age prediction. The next most important feature is **ethnicity_english**, followed by **silence_duration**, **jitter**, several MFCC-based descriptors, **num_pauses**, **hnr**, **shimmer**, **gender**, and f_0 -related pitch measures.

Overall, the ranking shows that the model does not rely on a single type of information. Instead, it combines features related to audio duration, silence and pause behavior, spectral characteristics, voice quality, pitch, and metadata. This result is consistent with the feature-group comparison, where the best performance was achieved when all feature groups were used together.

However, these importance values should be interpreted carefully. Permutation importance shows which features helped the model make predictions, but it does not prove that these features directly cause age differences. In particular, demographic variables such as **ethnicity_english** and **gender** may reflect patterns or imbalances in the dataset rather than direct age-related effects. Therefore, while these

Table 5: Selected top features by permutation importance.

Feature	Mean importance
aud_duration_audio	1.56 (approx.)
ethnicity_english	0.70 (approx.)
silence_duration	0.37 (approx.)
jitter	0.18 (approx.)
aud_mfcc_delta2_5_std	0.17 (approx.)
aud_mfcc_delta_5_std	0.11 (approx.)
num_pauses	0.11 (approx.)
aud_mfcc_6_mean	0.10 (approx.)

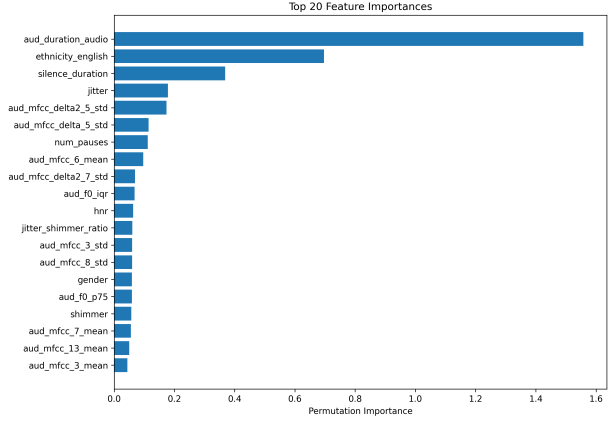


Figure 5: Top permutation feature importances for the final model.

variables improve predictive performance, they should be considered carefully in any broader or real-world application because of possible fairness and bias concerns.

6 Prediction Diagnostics and Error Analysis

Figure 6 shows the practical behavior of the final model. The true-versus-predicted scatter shows that predictions for younger speakers tend to lie closer to the diagonal than predictions for older speakers. This matches the age distribution in the dataset. Younger ages appear much more often, so the model has many more examples to learn from in that range. In contrast, predictions for older speakers are pulled more toward the middle age range. This is a common regression-to-the-mean pattern when extreme ages are underrepresented.

The uploaded prediction diagnostics support this interpretation. Some cases with moderate true ages are predicted reasonably well, but there are also substantial errors, including examples where very young or older speakers are predicted much closer to middle

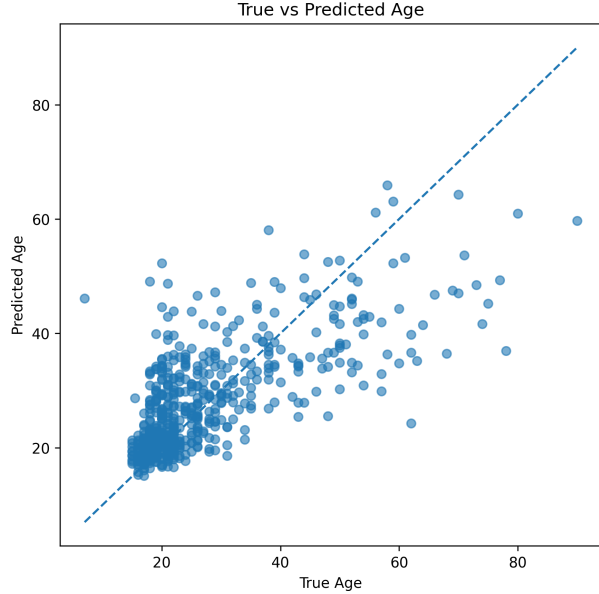


Figure 6: True versus predicted age for the final model.

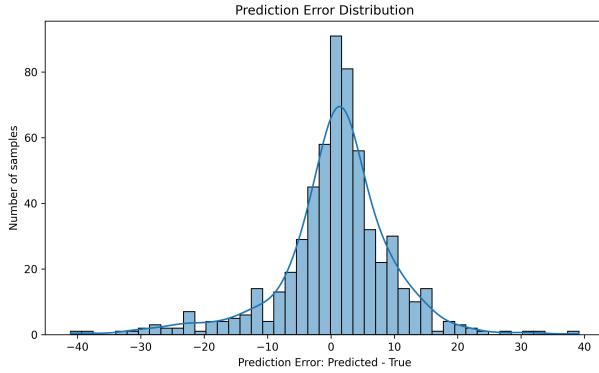


Figure 7: Distribution of prediction errors.

ages. These examples show that the model captures general age-related patterns, but still struggles on cases that are less common in the training data.

Figure 7 complements this analysis by summarizing the residuals. The error distribution is centered close to zero. This suggests that the model does not have a strong overall tendency to overpredict or underpredict. This is encouraging because it means the model is not mainly succeeding through a large one-direction bias. However, the distribution also exhibits noticeable tails, showing that some predictions are still far from the true age. Together with the scatter plot, these tails suggest that the main weakness of the final model is not overall bias. Instead, the problem is uneven reliability, especially in age ranges with relatively little data.

7 Discussion

Overall, the results show that HistGradientBoosting is the most effective model for this task. This makes sense because boosting methods often work well on tabular prediction problems with nonlinear relationships and feature interactions. Both are likely to be present in this speech dataset. ExtraTrees also performs strongly, which reinforces the broader conclusion that tree-based ensemble methods are a good match for age prediction from structured audio and metadata features.

The much weaker baseline result is also important. Because the baseline DummyRegressor ignores all input structure and simply predicts the average age, its inferior RMSE confirms that the feature space is genuinely informative. The models are not just using simple properties of the age distribution. They are using meaningful patterns in the input features to improve prediction. Hyperparameter tuning then strengthens this conclusion by improving the selected HistGradientBoosting model further and yielding the best final test accuracy.

The feature-group experiments add an important interpretive layer. They show that the full feature representation is best, that extracted audio descriptors carry especially strong signal, and that pitch and voice-quality variables alone are not enough for high-quality prediction. This means that age-related information is spread across several parts of the speech representation, including temporal structure, spectral information, and metadata. Finally, the diagnostic plots show why a single metric such as RMSE should not be interpreted alone. Even when overall accuracy is good, model behavior can still vary across different parts of the age distribution.

8 Limitations

The main limitation of this study is the imbalanced age distribution. The dataset contains many younger speakers but relatively few older speakers. As a result, RMSE may mask different levels of performance across age groups. In addition, the importance of demographic variables raises fairness concerns that should be examined explicitly before any broader deployment. Finally, feature importance shows predictive association, not causation. Therefore, the ranked variables should not be interpreted as direct causes of age-related differences in speech.

9 Conclusion

This study shows that speaker age can be predicted in a meaningful way from structured speech and meta-data features. It also shows that HistGradientBoosting is the strongest model among those tested. The model achieved the best initial cross-validation performance, improved further through hyperparameter tuning, and reached a final test RMSE of approximately 9.04 years, clearly outperforming the baseline. Feature-group and feature-importance analyses also show that the best results come from combining audio, temporal, voice-quality, and metadata information, rather than relying on one narrow type of cue. At the same time, the prediction diagnostics show that performance is more reliable for younger speakers than for older speakers. This reflects the imbalance of the dataset. Overall, the results are promising. However, future work should include age-bin analysis and fairness-aware validation to better assess robustness across the full population.

References

- [1] Jerome H. Friedman. Greedy Function Approximation: A Gradient Boosting Machine. *The Annals of Statistics*, 29(5):1189–1232, 2001.
- [2] Pierre Geurts, Damien Ernst, and Louis Wehenkel. Extremely Randomized Trees. *Machine Learning*, 63(1):3–42, 2006.
- [3] Guolin Ke, Qi Meng, Thomas Finley, Taifeng Wang, Wei Chen, Weidong Ma, Qiwei Ye, and Tie-Yan Liu. LightGBM: A Highly Efficient Gradient Boosting Decision Tree. In *Advances in Neural Information Processing Systems 30*, pages 3146–3154, 2017.
- [4] Scikit-learn developers. HistGradientBoostingRegressor documentation. *Scikit-learn documentation*, accessed 2026.
- [5] Damian Kwasny and Daria Hemmerling. Gender and Age Estimation Methods Based on Speech Using Deep Neural Networks. *Sensors*, 21(14):4785, 2021.